

H2GUARD · DESIGN DECISIONS REGISTER

H2Guard design decisions register

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DOCUMENT CONTROL

Document ID	HGD-DEC-001	Version	0.4
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2guard		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Amish accepted the cost overrun against the value-engineering target on 2026-10-03; row added to decisions made; value engineering section updated
0.3	2026-10-02	Amish Chadha	Decisions carried into the model, BOM and calculations; value engineering re-priced (USD 411); items to confirm for the drop rod, pressure switch and dry-contact relay
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 12 (HGD-DDR-003 accepted; head on a ceiling drop rod); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions gathered from the review note and the decision records; budget treated as a value-engineering target

H2Guard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The catalytic sensor's response time (t_{90}), power and hydrogen output, from its datasheet	R4 assumes a t_{90} of 30 s; R1's 1 % LFL resolution needs the output	HGD-CAL-001, F3
2	The sensor can is about 20 mm across and 17 mm tall	The standoff length sets its 2.2 mm gap above the disc	HGD-DDR-003, P4
3	The head box has its lid on a 110 x 90 face	The ports, gland and drop rod hole assume it	HGD-DDR-003, P3
4	The controller box's boss spacing and corner fixing holes	They set the mounting plate holes and the wall screws	HGD-DDR-003, P6, P8
5	The fan's body fits a 200 mm sleeve (about 180 mm across or less) and has fixing holes in its inlet face	The fan plate holds it by those holes	HGD-DDR-003, P10
6	The fan's curve gives at least 300 m ³ /h against about 60 Pa	R7 rests on an assumed curve	HGD-CAL-001, G3
7	The valve has two tapped holes underneath, 36 mm apart, and is rated for hydrogen with a stated seat leak rate	The bracket holes; the leaking-seat fault in Table 4 of HGD-CAL-001	HGD-DDR-003, P11; HGD-DDR-001, O5
8	The pressure switch can be set to about 3 Pa (a low-range air switch)	The fan inlet tap gives only 5.7 Pa at the continuous flow; a common 20 Pa minimum switch would prove flow only on boost	HGD-CAL-001, G6; decision 6
9	The floor flange, pipe nipple and conduit locknuts share one 1/2 in thread, and the ceiling takes the flange screws	The drop rod holds the head over the apparatus	HGD-CAL-001, E7; decision 2
10	The dry-contact relay's contact rating suits H2Bench's supply cut-off circuit	H2Bench switches its own supply through it	Decision 9

■ Value engineering

Value-engineering target: USD 265. Estimated cost of the constructable design: USD 411 (USD 146 over the target). The estimate comes from the 21-line bill of materials at indicative prices; the target is `budget_usd`, a hypothetical control target, not a limit. Two options are priced with quantity 0 and are not in the total: the oxygen sensor (USD 110) and the fan with its motor outside the air stream (USD 450, in place of the USD 55 fan). Main cost drivers and savings worth trying:

Amish accepted this overrun on 2026-10-03: the estimated cost of USD 411 against the USD 265 target (USD 146 over). Amish: "Cost over target - i accept all the cost variations and overruns". It stays reported against the target as an accepted overrun, and the savings below remain worth trying.

- The largest lines are the hydrogen-rated valve (USD 95), the exhaust fan with its grille, sleeve, shutter and hood (USD 55), the catalytic sensor (USD 35), the pressure switch and tubing (USD 30) and the controller modules (USD 25). Together they are about 60 % of the cost.
- Making the design constructable added USD 25: the made brackets and plates (USD 14), the 200 mm wall sleeve (USD 5), standoffs, screws and epoxy (USD 4) and the cup's push-in fitting and tube clips (USD 2).
- The decisions of 2026-10-02 added USD 122: the hydrogen-rated valve (USD 65 more than the brass valve), the pressure switch and tubing (USD 30), the ceiling drop rod (USD 11), the dry-contact relay and terminal (USD 6), the clear-lid controller box (USD 5) and more cable (USD 5).
- Savings worth trying: a printed controller mounting plate in place of aluminium sheet (about USD 3); buying the fan as a kit with its own shutter and hood; one combined controller board at TRL 4 in place of separate bought modules; and, where the room already has mechanical extract at high level, using it in place of the kit fan and grille (USD 63), which would bring the kit to about USD 348; and the brass valve at USD 30, if its maker states hydrogen compatibility and a seat leak rating and it passes the seat leak proof test (USD 65 less).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D9: catalytic plus metal oxide sensors (certified sensor as the upgrade); set points 10 % and 25 % LFL; normally closed valve; continuous ventilation with boost; inventory rule (R9); key-switch reset and one head per room; a university teaching lab first; independent hardware trip; 24 V DC with a certified supply	Amish: "i accept all your recommendations, go with them across all repos."	HGD-DDR-001, HGD-DDR-002
2026-09-25	Budget figure USD 265, now treated as the value-engineering target	Amish, same instruction; on 2026-10-01: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	HGD-DDR-002
2026-09-25	Timed escalation: a warning held 5 min closes the valve and latches	Amish, same instruction	HGD-DDR-002
2026-09-25	Head placement rule: a head directly above each likely leak point; offset limit set by test at TRL 4	Amish, same instruction	HGD-DDR-002
2026-09-25	R13 scope: the fan and grille wall openings are builder's work	Amish, same instruction	HGD-DDR-002
2026-09-25	H2Bench to reflect the inventory rule	Amish, same instruction	HGD-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	HGD-DDR-003 (changes accepted on 2026-10-02, below)
2026-10-02	Design for construction accepted: the changes P1 to P11, as made; the head position is settled separately (decision 2)	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, Table 1
2026-10-02	The head hangs from the ceiling on a drop rod directly over the apparatus; wall mounting is allowed only if the TRL 4 plume test shows a trip at the wall position (the wall position puts the ports 260 mm from the apparatus centre against a plume radius of about 141 mm)	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, A1
2026-10-02	Fan motor in the sleeve accepted for the supervised teaching prototype; any room used without close supervision gets a fan with its motor outside the air stream	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-003, A2
2026-10-02	An oxygen sensor is added, as a standard kit option, whenever inert gas cylinders share the room	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O2
2026-10-02	Set points and the inventory rule come from the stricter of the national code and ISO 26142, and from the host institution's safety office where its	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O3

Date	Decision	Decided by	Record
	rules are stricter; 10 % and 25 % LFL stay the defaults		
2026-10-02	A differential pressure switch at the fan proves airflow	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O4
2026-10-02	The valve is rated for hydrogen with a stated seat leak rate; the low-cost brass valve is accepted only if its maker states hydrogen compatibility and a seat leak rating and it passes a seat leak proof test at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	HGD-DDR-001, O5
2026-10-02	The 1 % full-span bump test trips the system, as planned; the pass level is set once the cup delivery factor is measured	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-25
2026-10-02	A dry-contact relay output that opens on a trip or loss of power is added, for H2Bench to cut its own supply	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-25
2026-10-02	The render layout is kept, with a caption saying the head and valve are drawn beside the controller	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 1
2026-10-02	A clear-lid IP65 controller box, added to the BOM at its next revision	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 2
2026-10-02	Leaving the fan, grille, sounder, beacon and supply out of the product renders accepted	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26, item 3
2026-10-03	Cost overrun accepted: the estimated cost of USD 411 against the USD 265 target (USD 146 over)	Amish: "Cost over target - i accept all the cost variations and overruns"	REVIEW.md , session 2026-10-03